

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – EXPERIMENTS

Course Code:	CSA0620	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Articulation (BL4, BL5)	Assessment:	Assessment Tool 1 – Experiments
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:	Sairaj C	Reg. No.:	192411018
Section / Batch:	2024	Date:	01-08-2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Write the algorithm and execute the code for the given experiment.
- Follow a well-structured, systematic approach to design and implement an optimal solution.
- Provide insightful analysis with accurate validation, supported by clear documentation including diagrams, code, and results.

Question Type	Focus Area	Questions
Experiments	Design and Code for Divide and Conquer Algorithms: Merge Sort, Quick Sort, Binary Search, and Matrix Multiplication	Q1

SECTION A – Experiments

Q1. Analyze the impact of input size on recursion depth in Merge Sort by implementing the algorithm and recording recursion levels for datasets of varying sizes. Plot recursion depth versus input size and interpret the growth pattern. Examine how divide and conquer reduces problem size at each level. Justify the relationship between recursion depth and time complexity.

Code of Conduct

I Sairaj C (192411018) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sairaj C

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Language: Python C C++ Java

QUESTIONS

Q1
Quick Sort
10 marks

1/1 answered

Q1 Quick Sort
10 marks

Evaluate Quick Sort using hybrid approaches by combining insertion sort for small subarrays. Record execution time for different threshold values. Analyze performance improvements. Interpret the results and justify the effectiveness of hybrid sorting techniques.

Your Code

```

1 import random,time
2 def insertion_sort(arr, low, high):
3     for i in range(low + 1, high + 1):
4         key = arr[i]
5         j = i - 1
6         while j >= low and arr[j] > key:
7             arr[j + 1] = arr[j]
8             j -= 1
9         arr[j + 1] = key
10
11 def partition(arr, low, high):
12     pivot = arr[high]
13     i = low - 1
14     for j in range(low, high):
15         if arr[j] < pivot:
16             i += 1
17             arr[i], arr[j] = arr[j], arr[i]
18     arr[i + 1], arr[high] = arr[high], arr[i + 1]
19     return i + 1
20
21 def quick_sort(arr, low, high):
22     if low < high:
23         p = partition(arr, low, high)
24         quick_sort(arr, low, p - 1)
25         quick_sort(arr, p + 1, high)
26
27 arr = [random.randint(1, 100) for _ in range(100)]
28 start = time.time()
29 quick_sort(arr, 0, len(arr) - 1)
30 end = time.time()
31 print("Time taken: ", end - start)

```

Input

stdin input

Output

```

Threshold | Time (s)
0 | 0.000000
10 | 0.015633
20 | 0.000000
50 | 0.000000
100 | 0.000000

```

RUBRICS FOR CALCULATION

Experiments					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding ; unable to integrate concepts
Experimental Design	30%	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20%	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15%	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10%	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
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Understanding of Concepts	25%				
Experimental Design	30%				
Use of Tools	20%				
Analysis of Results	15%				
Documentation	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____